

Deep ABD adjudication

the $k = 10/11/12$ breaks are REAL;
mesh repair at the 2^k anchors fails

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Abstract

r445 found selected $k = 10$ (107 sign-flips from $n = 3788$), $k = 11$ (839 flips, B_w explosion), $k = 12$ (abort at $n = 12737$). The Lean mincut needs infinitely many Selected windows to be ABD-living. r397 proved $a_k \rightarrow \infty$ and $\Delta_k \rightarrow 0$, not chain positivity. This note re-runs the $\tilde{\mu}$ -chain at $k = 10$ in mpmath (dps 40 and 60) on the r445 float64 atoms. The first flip stays at $n = 3788$ with $\eta = -7.938908159491412 \cdot 10^{-14}$, identical at both dps. The breaks are *not* float64 recurrence artefacts. Legal m -neighbours of the 2^k anchors are likewise ABD-dead; the last observed living window is k_z136 . No RH claim.

Status. No new SATZ. Propositions 2–3 are a named infra census. Ausgang REAL / MESH_DOES_NOT_REPAIR / LAST_LIVE_KZ_136 / K12_ETA_UNDERFLOW / SLICE_FLOOR_STANDS / COFINAL_ABD_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Pivot anatomy

ABD.ok requires the $\tilde{\mu}$ -chain to reach depth N with every $h_n > 0$. The recorded pivot is the signed mass $\eta = P - N$ at step *start*, $P = \sum w q^2$, $N = \sum v p^2$, $\text{rel} = |\eta|/(P + N)$.

Proposition 1 (Float first flips). *On the selected sequence the float64 chain records: $k = 10$ first flip $n = 3788$, $\eta = -7.87 \cdot 10^{-14}$, $P \approx N \approx 3.44 \cdot 10^{-8}$, $\text{rel} = 1.14 \cdot 10^{-6}$; $k = 11$ first flip $n = 10581$, $\eta = -7.64 \cdot 10^{-16}$, $\text{rel} = 4.81 \cdot 10^{-8}$; $k = 12$ first flip $n = 12654$, $\eta = -1.01 \cdot 10^{-22}$, $\text{rel} = 4.66 \cdot 10^{-14}$, then $\eta = 0$ abort at $n = 12737$ (class *ETA_UNDERFLOW*). $k = 8, 9$ have 0 flips ($\min |\eta| = 2.6 \cdot 10^{-14} / 5.5 \cdot 10^{-12}$).*

Six-digit cancellation at $k = 10$ still leaves about ten float64 digits. That is *not* an underflow of η itself; it is small enough that a recurrence-rounding story was live (r427 ward: pivots $\sim 10^{-8}$ at $N \geq 4071$).

2 mpmath adjudication

Atoms stay float64 (the construction). The three-term recurrence is re-run at high dps. In-window start- η at $k = 5$ ($N = 96$) agrees with float to 10^{-14} relative (the post-window extra step is not an ABD row).

Proposition 2 (REAL). *The $k = 10$ chain at dps 40 (full, 503s) flips first at $n = 3788$, $\eta = -7.938908159491412 \cdot 10^{-14}$, $\text{rel} = 1.150853124516389 \cdot 10^{-6}$, $n_{\text{flip}} = 115$, $\text{pos_ok} = \text{false}$. The same first pivot at dps 60 (to $n = 3793$, 428s) is bit-identical. The sign is stable under a dps raise. Hence the r445 break is REAL relative to the float64-atom construction: it is not a float64 recurrence artefact.*

$k = 11$: the first flip is the same near-null class at tighter rel; the B_w explosion is signed $\rho = f_b^2/\eta$ after many flips, not a second phenomenon. A full $k = 11$ mpmath chain to $n = 10581$ is outside the 45 min budget (larger atoms, $\sim 3\times$ steps). $k = 12$ dies by ETA_UNDERFLOW: float hits exact $\eta = 0$ on the same near-null (not overflow, not a division exception).

3 Mesh

r397 leaves room in the m_k choice at fixed anchors $a_k = 2^k$. pp_kz rounding does not move $k = 10, 11, 12$ (2^k sits exactly on the U -grid).

Proposition 3 (Mesh does not repair). *Every tested legal neighbour of the $k = 10$ anchor ($k_z193..201$) and of the $k = 11$ anchor ($k_z337..341$) is ABD-dead in float. A denser sample $k_z137..210$ is likewise dead. The last observed living window is k_z136 ($N = 1641$); k_z137 ($N = 8300$) already flips. No small m -variation at the 2^k anchors restores ABD-living.*

This is a finding about the construction, not a rounding bug. The r397 cofinality theorems ($a_k \rightarrow \infty$, $\Delta_k \rightarrow 0$) do not imply ABD-cofinality. A revised sequence that stays in the living band would need a different window family (not a local m -tweak of $a_k = 2^k$). Whether infinitely many living windows exist at all is open; the empirical living band ends at k_z136 in every sample we ran.

The r445 ABD-ok slice $k = 5..9$ (live $k = 8$) and its floor fit are unchanged ($\delta_\infty = +0.02741$, $\Delta\text{AIC} = 5.11$). There is no new deep δ -point.

Dead χ_{3-15} : $\delta = -0.023181$ (control). $k = 5$ determinism: run1 = run2.

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis. The identities are a finite recurrence on a named atom construction. The frequently-quantifier gap (ABD-living of the Selected sequence) stays open.

References

- [1] Bordered chain / $B_w = S_{N-2} + 5/7$.
- [2] Selected sequence; $a_k \rightarrow \infty$, $\Delta_k \rightarrow 0$.
- [3] Campaign audit: tiny pivots at depth.
- [4] Deep builder; $k = 10/11/12$ not ABD-living (float census).